# Who Does the AI Work For? Negotiating an AI Agent's Role Between Older Adults and Family Caregivers in Bangladesh

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**ACM Reference Format:**
Anonymous Author(s). 2026. Who Does the AI Work For? Negotiating an AI Agent's Role Between Older Adults and Family Caregivers in Bangladesh. 1, 1 (September 2026), 28 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

## 1 Introduction

A missed dose can create a difficult choice: should an older adult manage it independently, or should a family member step in? Family involvement can support medication routines [92], yet the same checking can shift privacy, responsibility, and control [24, 115]. This tension matters especially in many Global South households, where medication work often extends across relatives, shared devices, and intermediated technology use [2, 98]. Reminding, purchasing medicines, interpreting prescriptions, and checking doses can therefore be part of everyday family care rather than separate acts of assistance. Medication adherence is more than a problem of remembering. It is also a continuing negotiation over who notices, who acts, and who takes responsibility when a routine breaks down.

Family support can preserve older adults' agency, but it can also redraw its boundaries. Older adults may rely on others for assistance while still wanting control over how support and health information are shared [13, 24, 30, 50]. Interdependence and aging scholarship similarly shows that receiving assistance need not diminish agency because people can rely on others while retaining a meaningful role in how support is arranged [13, 30, 33]. Yet those boundaries can change across situations: checking may communicate care at one moment and feel like unwanted oversight at another [24, 115]. Designing for medication adherence therefore requires support for these shifts without assuming that either independence or family involvement should always take priority.

This negotiation becomes harder when medication technology can act before anyone explicitly asks it to. Mixed-initiative systems have long considered when computational systems should take initiative, while newer proactive agents can intervene in communication or propose actions on a person's behalf [45, 52]. In medication care, an AI agent might notice an unconfirmed dose and ask whether a relative should be involved. If the agent keeps the missed dose private, the older adult retains control, but the caregiver may remain unaware. Involving the family can bring support, while also expanding who knows about the dose and who becomes responsible for responding. The AI agent therefore becomes a participant in deciding how responsibility moves between older-adult control and family involvement.

We call this relationship **allegiance**: whose interests and authority the AI agent gives priority to at a particular moment. Allegiance is not simply a permission setting or a fixed assignment to one user. The same older adult may want private support during an ordinary routine, welcome shared involvement after uncertainty, and resist family intervention in another situation. Because these preferences can shift, allegiance must be visible and revisable rather


Manuscript submitted to ACM

than hidden in configuration. The design problem is not to align the agent once, but to let people inspect, negotiate, and change whom it serves as circumstances and relationships change.

Care technologies have established that family visibility can support coordination while also redistributing privacy, responsibility, and power. Awareness displays have helped relatives follow older adults' activities and coordinate care [22, 82, 94], and collaborative systems have shown how families can participate in person-centred care [81]. Prior studies also show that shared visibility can create new obligations and tensions over who can access information or act on it [24, 111, 115]. These studies make family participation visible as part of care, not as an external interruption. They leave a further question when the system can itself propose that responsibility move from one person to another.

Research on agent alignment and ways to question automated decisions provides a second foundation for this problem. Alignment research has examined how agents should cooperate with, obey, or represent human objectives [32, 39, 79]. Work on agents that must account for multiple people's interests further shows how difficult it can be to decide whose interests should take priority [62]. Research on questioning AI decisions, exposing system uncertainty, and revising consent shows how automated actions can remain open to scrutiny and change [7, 28, 50, 71]. Together, these traditions explain why an agent's authority must remain visible and open to revision. What remains unresolved is how that authority is negotiated when an older adult and caregiver both have legitimate, but sometimes different, expectations of what the agent should do.

We study allegiance as an ongoing negotiation rather than a fixed permission or caregiver-access setting. We conducted a two-phase qualitative household study in Bangladesh with 17 older adults and eight family caregivers. Phase 1 examined medication practices and the ways families already shared responsibility. We then developed and deployed a medication-support prototype for two weeks that reminded older adults, recorded medication activity, stated whom it was serving, and requested permission before involving a family member. Post-deployment interviews examined how participants responded to these interactions. Accounts of deployed features were kept separate from judgments about additional mechanisms that were described but not built. Across the study, we analyse when participants treated the agent as a **tool** for the older adult, a **coach** shared across the care relationship, or an **advocate** for greater family involvement. We also examine what made movement among these roles accepted, resisted, or reversed.

Together, these phases trace how responsibility is negotiated before the agent arrives, how allegiance is negotiated after it enters care, and what those negotiations require from design. The study asks:

**RQ1 (Formative).** How do Bangladeshi older adults and family caregivers distribute and negotiate medication work, and how do they understand responsibility for that work before an agent enters the household?

**RQ2 (Interaction).** When an agent joins this care network, how do older adults and caregivers negotiate its allegiance, and when are shifts between older-adult control and family involvement accepted, resisted, or reversed?

**RQ3 (Design).** What makes the agent's roles as tool, coach, or advocate legitimate to older adults and caregivers, and what design implications follow for making shifts among these roles visible, revocable, negotiable, and dignity-preserving?

This work makes three contributions:

- **Empirical.** This study shows how Bangladeshi older adults and family caregivers negotiate medication responsibility and family involvement across formative and post-deployment accounts.
- **Methodological.** We use a caregiver-inclusive household approach that treats caregivers as participants in their own right and preserves divergent accounts of shared medication work.
- **Design.** The study identifies directions for shared-care agents that keep shifts in allegiance visible, negotiable, and reversible.

For shared-care agents, whom the system serves cannot remain a hidden configuration choice.

## 2 Related Work

### 2.1 Medication Management as Shared Care Work

Medication adherence is not only about remembering a dose; it is also about coordinating routines, information, and responsibility. Reminder apps can help people notice a dose. Yet studies of older adults and people managing several chronic conditions show that adherence also depends on treatment complexity, daily habits, and how well technology fits existing routines [27, 35, 107, 108]. Medication work can also include interpreting bodily changes, managing health information, and coordinating treatment across people and settings [8, 9, 54, 85, 119]. Conversational systems and studies of family health routines extend this view by showing that self-management may involve relatives and care partners rather than one person acting alone [74, 92, 102, 120]. This makes medication management a useful setting for studying how care work is shared.

Shared medication work is easier to understand when care is treated as an ongoing practice rather than a sequence of isolated choices. Practice-oriented HCI describes technology use as part of everyday activity, while care theory emphasizes repeated adjustment to changing needs [61, 80, 112]. Research on invisible work likewise shows that patients and caregivers spend substantial effort organizing information, monitoring changes, and keeping care moving [8, 47, 85, 106]. Feminist and relational approaches add that care is shaped by responsibility, responsiveness, and relationships, not only by individual choice [10, 43, 44, 72]. Once medication management is viewed this way, the next question is who gets to decide how that work is divided.

Older adults can rely on others without giving up agency. HCI research challenges accounts that treat older adults as passive recipients, reluctant users, or people whose needs can be inferred from age alone [11, 41, 48, 58, 59, 64, 65]. Interdependence research instead shows that people may depend on others while retaining meaningful control over how help is arranged [13, 30, 78, 101]. Studies of technology adoption and home-based care similarly describe agency as shaped through relationships, social position, and information practices [56, 86]. The issue is therefore not whether help exists, but whether the older adult still has a meaningful role in shaping it.

Family care makes this tension visible because support and control can move together. Care-network research has long shown that older adults, relatives, and care workers coordinate information and responsibilities across people [22, 63, 82, 94, 118]. More recent work on dementia care, home monitoring, and medication support shows that family involvement can improve coordination while also creating disagreements over access, monitoring, and decision-making [24, 36, 73, 76, 81, 115]. These studies establish that family support can be valuable without assuming that older adults and caregivers always want the same thing.

*What remains unclear is what happens when the system itself proposes a change in this division of work. Prior research explains reminders, family coordination, and relational agency, but gives less attention to how older adults and caregivers respond when an autonomous agent decides that another person may need to become involved.*

### 2.2 Family Visibility, Privacy, and Dignity in Later-Life Care

Family visibility can support care, but it also changes who knows, who acts, and who feels responsible. Awareness displays and connected care tools can help relatives follow an older adult's daily life [22, 82, 94, 118]. Yet monitoring and information-sharing studies show that the same visibility can create conflict over privacy, data ownership, and expectations to intervene [23, 24, 66, 87, 111, 115]. Research on shared-device privacy reaches a similar conclusion: access rules are often negotiated between people rather than set once by one user [50, 122]. This shifts the problem from whether information should be shared to how that sharing changes relationships.

Those relational changes also affect dignity. Older adults may reject technologies that signal decline, and visible assistive devices can affect social standing or the desire to preserve face [17, 20, 68, 73]. Work on domestic robots similarly shows that older adults weigh dignity, autonomy, and the kind of relationship a system appears to create [21]. Goffman's account of self-presentation helps explain why the social meaning of assistance can matter alongside its practical benefit [34]. A system can therefore preserve formal choice and still feel diminishing if it repeatedly presents a person as dependent.

Because these effects are relational, caregiver perspectives matter without replacing the older adult's account. Caregivers often move among several roles and perform work that systems do not fully capture [47]. Studies of home care and security decisions show that caregivers and care recipients can describe the same arrangement differently, including situations where apparently shared decisions leave the older adult with little practical choice [24, 56, 66, 76]. Family information-sharing can likewise improve coordination while creating new privacy concerns for the person receiving care [115]. These differences lead directly to the question of whose interests a system should prioritize when family members disagree.

Autonomy, privacy, dignity, and caregiver coordination each explain part of that problem, but none explains all of it. Autonomy concerns whether the older adult retains meaningful control; privacy concerns what becomes visible and to whom; dignity concerns how support affects social standing; caregiver research explains how responsibilities are distributed. These concepts help identify the stakes. They do not by themselves specify whose interests an autonomous agent should prioritize at a particular moment.

*The gap is therefore not a lack of theories about family care. It is that existing theories do not fully explain how an agent should move between older-adult control and family involvement, or how people judge whether that shift is appropriate in a particular situation.*

### 2.3 Family-Mediated Technology Use in Bangladesh

In Bangladesh, family involvement is not an added layer around care; it is often part of how care is organized. Research on ageing in Bangladesh and South Asia describes limited formal eldercare, strong reliance on relatives, and changing intergenerational arrangements [5, 12, 37, 51, 77, 90]. An asset-based perspective cautions against describing this only as a lack of formal services, because households and communities already hold practices through which support is provided [60]. This means that an agent entering the household also enters an existing network of family responsibility.

Technology use in that network is often shared as well. Studies in Bangladesh document collaborative phone use, privacy negotiation on shared devices, and privacy risks that arise through ordinary repair and household practices [1–4]. Research on intermediated technology use shows that one person may operate, explain, translate, or manage a digital resource for another [33, 98]. Such support can be a practical way of gaining access rather than evidence that the person receiving help has no agency. Work with rural women in Bangladesh further shows that technology can both reinforce unequal relations and provide ways to exercise agency within them [110]. Shared use therefore makes authority a household issue, not only an interface issue.

Global South HCI explains why these household arrangements should not be measured against an individual-user model by default. Postcolonial computing and critiques of universal design assumptions show how history, infrastructure, and power shape technology use [26, 49]. Structural approaches to digital care similarly show that technology can redistribute existing care work rather than simply add a neutral service [53]. Research on sustainable ICTD in Bangladesh, explainable AI in Global South settings, and Global South perspectives on AI governance further shows that systems are interpreted through local institutions, workflows, and stakeholder positions [84, 88, 96]. Studies of rural clinical AI and voice assistants add that technical adoption does not guarantee a good fit with everyday practice [89, 114]. These findings make the household context central to any account of agent authority.

Existing Bangladesh and ICTD work has therefore established shared devices, mediated use, relational privacy, and household power as important design concerns [1–4, 33, 98, 110]. Across much of this work, however, the technology remains a tool, information system, or shared device rather than an agent that decides when another person should become involved [26, 49, 53, 84, 88, 96, 114]. That difference matters because initiative changes the question. Families must negotiate not only who can use the system, but whose interests should guide what it does.

*What remains underexplored is how older adults and caregivers in such households negotiate authority when an autonomous agent begins acting within existing family care. Our study examines this gap in Bangladesh without assuming that receiving family help means less agency, or that family-based care means everyone will agree.*

### 2.4 From Agent Initiative to Allegiance

An agent becomes part of the care relationship when it can act before anyone explicitly asks it to. Mixed-initiative research established that systems may balance user control with computational initiative, while newer proactive agents can suggest actions, communicate on a user's behalf, or begin support without a direct command [45, 52, 70]. Research on plan disclosure and AI decision power shows that people care about when an agent acts, how much authority it has, and whether its intended action is visible beforehand [42, 55]. Trust research similarly argues that appropriate reliance depends on understanding what a system can and cannot be trusted to do [57, 67]. Initiative therefore raises a question of authority, not only capability.

Contestability and oversight research explains how that authority can remain open to challenge. People need ways to question automated actions, intervene, and understand important system boundaries and decision points [6, 7, 25, 28]. Responsibility research further shows that autonomous action can make it harder to determine who should answer for a decision or its consequences [99]. In family care, this problem becomes sharper because involving a caregiver can change privacy and responsibility at the same time. The next question is whose interests should guide that action.

Alignment research provides one answer by asking what an agent should serve. Foundational work distinguishes instructions, preferences, interests, and values, while formal models show that obedience and benefit do not always point in the same direction [32, 38, 39, 79]. More recent pluralistic approaches recognize that agents may face several legitimate values, communities, or principals rather than one user with one stable preference [29, 62, 105, 121]. This work makes the question "for whom?" explicit. Much of it, however, remains formal, conceptual, or model-level rather than an account of how related people negotiate an agent's priority during everyday care.

Care technology reaches the same problem from the opposite direction. Multi-stakeholder systems already involve older adults, relatives, professionals, and caregivers, but many focus on awareness, monitoring, information sharing, or coordination [22, 24, 63, 81, 82, 94, 111, 115, 118]. Conversational systems can also support medication management or communication between older adults and care partners or providers [74, 117]. These systems show that several people can have legitimate interests in the same care process. What they offer less evidence about is how a household responds when the agent itself proposes that responsibility move from one person to another.

We use *allegiance* to name that specific interaction problem: whose interests and authority the agent prioritizes at a particular moment. Allegiance is narrower than alignment and different from a permission setting. Autonomy asks whether the older adult retains control; privacy asks what others can see; consent asks whether an action is allowed; alignment asks which instructions, values, or principals guide the agent. Allegiance focuses on how those concerns meet when more than one person has a legitimate claim over what the agent should do. An older adult may want private support in one situation, shared support in another, and no family involvement in a third. The priority can change.

*The central gap is empirical: plural-alignment research identifies the problem of multiple principals, while family-care research documents multiple stakeholders. We still know little about how older adults and caregivers negotiate an agent's changing priority between them during everyday use, or what makes such a shift accepted, resisted, or reversed.*

### 2.5 Making Allegiance Visible, Revisable, and Negotiable

A changing allegiance is only meaningful if people can notice and question the change. Contestable-AI research argues that automated decisions should remain open to human challenge, while work on explanations and oversight shows the value of making plans, limits, and intervention points clear [6, 7, 25, 28, 42, 52]. For a shared-care agent, this means more than explaining why a reminder appeared. People also need to know when the agent is about to involve someone else and what that involvement will change.

That visibility must be paired with the ability to change one's mind. Research on distributed and interactive consent argues that decisions involving several people require permission that can be discussed, updated, and withdrawn [71, 83, 104, 109, 122]. Studies with older adults similarly show that privacy boundaries change with circumstances rather than remaining fixed after setup [50]. Voice interfaces add another difficulty because a short verbal agreement can become routine without remaining meaningful [100]. Rather than treating caregiver access as permanent, shared-care agents therefore need ways for people to reconsider who is involved and under what conditions.

Making role changes visible can also create social costs. Research on stigma, dignity, and self-presentation shows that visible assistance can signal dependence or affect social standing [17, 20, 21, 34, 68]. Invisible-work research adds that making care work visible can change the expectations surrounding that work [85, 106]. Relational and feminist ethics therefore caution against treating transparency as automatically beneficial [10, 43, 44]. The design challenge is to make a change clear enough to question without repeatedly presenting the older adult as incapable.

Even small interface cues can shape these relationships. Research on nudging and persuasive design shows that prompts take meaning from the practices in which people receive them [18, 31]. Gamification research likewise cautions that points and streaks are not neutral mechanics, while studies of communication and family fitness show that shared metrics can become signals of closeness, obligation, encouragement, or loss [19, 40, 46, 69, 91, 97]. Self-determination research connects interactive systems to autonomy, competence, relatedness, and motivation, while warning against using these concepts as generic explanations for engagement [14, 95, 113, 116]. The broader lesson is that an agent's signals about responsibility may shape family relationships as well as individual behavior.

These relational shifts also affect how the problem should be studied. Research on reflexive thematic analysis stresses that qualitative interpretation should make analytic choices explicit rather than treating coder agreement as the only sign of quality [15, 16, 93]. Work on positionality and reflexivity in HCI similarly argues that researchers should account for how their standpoint shapes interpretation [75, 103]. This matters when older adults and caregivers may give different accounts of the same care episode. Keeping those accounts distinct allows disagreement to remain evidence rather than being averaged into one household view.

*What remains is a design and methodological gap. We need to understand how an agent's changing allegiance can be made clear, open to withdrawal or reversal, and negotiable without undermining dignity. We also need to study those shifts through the separate accounts of the people affected by them. This is the gap addressed by our caregiver-inclusive household study and by our analysis of the agent as tool, coach, or advocate.*

# 3 Method

## 3.1 Study Design

We conducted a two-phase qualitative household study in Bangladesh from January to June 2026. The study examined how older adults and family caregivers organized medication work within the household and how those arrangements shaped their experiences with a medication-support prototype. All study sessions took place in participants' homes and were conducted in Bangla.

**Phase 1** consisted of formative interviews, contextual observation, and medication-routine walkthroughs. We examined how medicines were stored and taken, how responsibilities were distributed among household members, how routines were disrupted, and how relatives participated in reminding, checking, purchasing, or otherwise supporting medication use. We conducted a formative analysis of Phase 1 before developing the prototype so that its design reflected practices documented in participating households.

We then developed and installed a medication-support prototype informed by the Phase 1 findings. Participants used the prototype for two weeks. The deployed system provided medication reminders and records, stated whom it was serving, and could ask whether a family member should be involved when a scheduled dose remained unconfirmed.

**Phase 2** consisted of post-deployment interviews conducted after the two-week deployment. These interviews examined participants' reported experiences with deployed features and their reasoning about several additional design mechanisms that had been specified but were not implemented. We distinguished these forms of evidence throughout the study: accounts of deployed features are treated as reported experiences of use, whereas responses to unimplemented mechanisms are treated as judgments about proposed designs.

We treated the household as the primary context of medication management because relatives often participated in purchasing medicines, reading labels, organizing doses, providing reminders, and checking whether scheduled doses had been taken [106]. Where such coordination occurred, we recruited an involved family caregiver as a participant in their own right. When an older adult and caregiver described the same household event differently, we retained both accounts rather than reconciling them into a single version.

## 3.2 Participants and Recruitment

We recruited participants through snowball sampling, beginning with the research team's social networks and continuing through referrals from friends, colleagues, and community groups. Approximately 30 households were approached. Participation was voluntary and unpaid.

Older adults were eligible if they were at least 65 years old and took medication daily. Family members were eligible if they performed medication-related work for an older adult living in the same household.

The final sample comprised 25 participants: 17 older adults and eight family caregivers. Phase 1 included 16 older adults and eight caregivers, while Phase 2 included 17 older adults and seven caregivers. Sixteen older adults and seven caregivers participated in both phases; one older adult participated only in Phase 2, while one caregiver participated only in Phase 1.

Older adults were 65–80 years old (mean = 69.4; median = 68), including nine women and eight men. Eleven needed large text, including two who could not read, while nine described low or very low smartphone comfort. Five reported taking one medicine daily, three reported eight or more, and one reported thirteen medicines across three dose times.

The eight caregivers included four women and four men. Six reported their age, ranging from 20 to 31 years. Their relationships to the older adults included daughters, sons, a daughter-in-law, and grandchildren. Tables 1 and 2 report participant-level demographic, medication, device, and caregiving information.

Table 1. **Older adults taking daily medication (n = 17).**

| ID | Age | Gender | Daily medicines | Dose times/day | Deployment device |
|---|---|---|---|---|---|
| P01 | 72 | M | 4 or more | 4 | Own smartphone |
| P02 | 65 | M | 1 | 1 | Household smartphone |
| P03 | 66 | F | 3 to 4 | 2 | Own smartphone |
| P04 | 69 | M | 4 | 3 | Own smartphone |
| P05 | 68 | F | 1 | 1 | Household smartphone |
| P06 | 67 | M | 1 | 1 to 2 | Household smartphone |
| P07 | 71 | F | 3 to 4 | 2 or more | Own smartphone |
| P08 | 67 | F | 2 to 4 | 2 | Household smartphone |
| P09 | 68 | F | 3 to 4 | 3 | Household smartphone |
| P10 | 78 | M | 2 | 3 | Own smartphone |
| P11 | 65 | M | 5 to 6 | 3 | Own smartphone |
| P12 | 71 | F | 1 | 1 | Own smartphone |
| P13 | 66 | F | 1 | 1 | Own smartphone |
| P14 | 66 | F | 5 | Not recorded | Not recorded |
| P15 | 80 | M | 8 to 10 | 3 | Not recorded |
| P16 | 76 | F | 13 | 3 | Not recorded |
| P17 | 65 | M | 8 | 2 | Own smartphone |

Table 2. **Family caregivers (n = 8).**

| ID | Age | Gender | Relation | Older adults supported | Daily medicines managed |
|---|---|---|---|---|---|
| C01 | Not recorded | F | Daughter | Mother and father | 3 to 4, and 2 |
| C02 | Not recorded | F | Daughter | Mother and grandfather | 2 or more, and 3 to 4 |
| C03 | 21 | M | Son | Father and mother | 3 to 4, and 2 |
| C04 | 31 | M | Son | Mother | 2 |
| C05 | 20 | F | Daughter-in-law | Father-in-law and mother-in-law | 4 to 5 |
| C06 | 22 | F | Granddaughter | Grandmother | 5 to 6 |
| C07 | 25 | M | Grandson | Grandfather | 10 |
| C08 | 25 | M | Grandson | Grandmother and mother | 5 to 6 |

### 3.3 Phase 1: Formative Household Study

Following prior situated-practice work [9, 61], Phase 1 examined medication practices before participants encountered the prototype. We used separate semi-structured interview protocols for older adults and family caregivers; the complete protocols are provided in **Appendix A1** and **Appendix A2**, respectively.

The interviews covered daily medication routines, medicine storage, disruptions, recent missed or delayed doses, reading and interaction conditions, family involvement, responsibility sharing, technology use, and what happened when the person who usually provided support was unavailable. Rather than asking only about general preferences, we asked participants to reconstruct recent episodes and demonstrate relevant parts of their routines in the places where medication work occurred [9, 61].

The caregiver protocol focused on caregivers' own medication-related work, including how responsibilities developed, how they checked whether medicines had been taken, and what happened when they were unavailable [106]. We distinguished caregivers' accounts of their own activities from their interpretations of the older adult's experience and from jointly described household events.

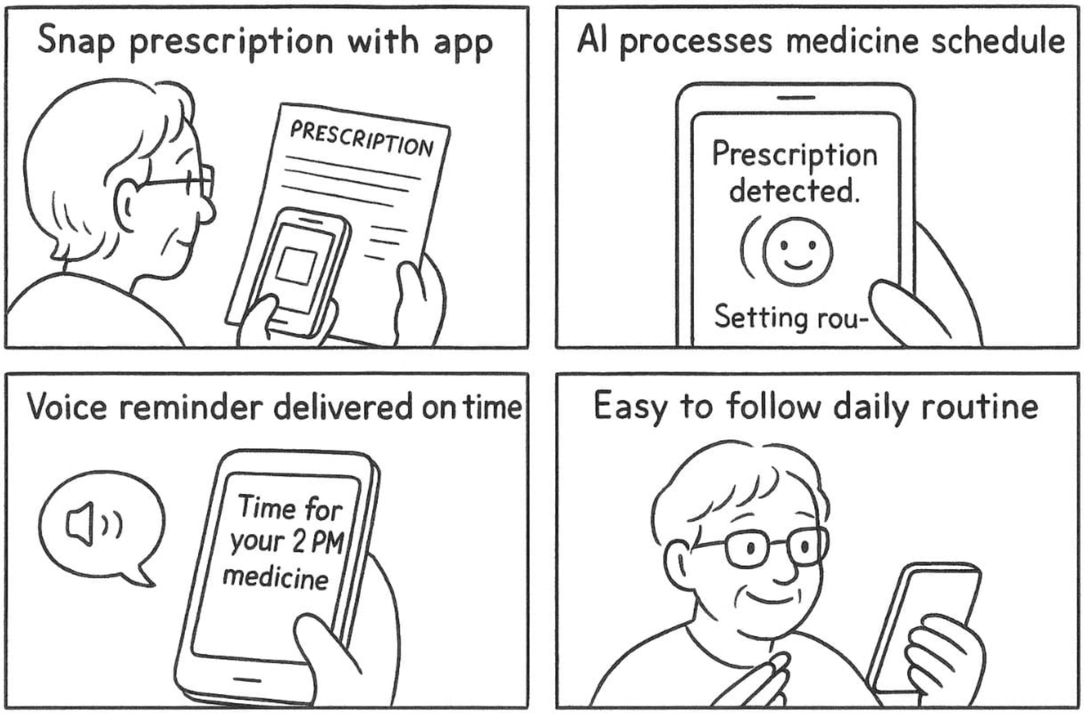


Fig. 1. Four-panel storyboard of the design concept: capturing a prescription, the schedule being set from it, a spoken reminder at dose time, and a routine the person can follow.

Each session followed the same general sequence: consent, observation of the immediate setting, semi-structured interviewing, a medication-routine walkthrough, and a closing design reflection. With permission, sessions were audio-recorded in Bangla. Researchers also documented medicine storage, relevant lighting and noise conditions, medication-related artifacts, and household members involved during demonstrations.

Phase 1 supports claims about existing medication practices and the requirements that informed prototype development. It provides no evidence about prototype use.

### 3.4 Formative Analysis and Prototype Development

We conducted a formative analysis of the Phase 1 material before prototype development. The analysis identified requirements concerning timely reminders, medicine and timing information, reduced dependence on manually consulting a schedule, family awareness of unconfirmed doses, simplified interaction, readable presentation, and opportunities for family involvement. After Phase 2, the Phase 1 material was considered again as part of the full qualitative analysis described below.

Figure 1 shows how these requirements informed the initial design concept. The concept included prescription capture, schedule creation, spoken reminders, and a medication routine organized around those reminders. Prescription capture was specified in the concept but was not implemented in the deployed prototype.

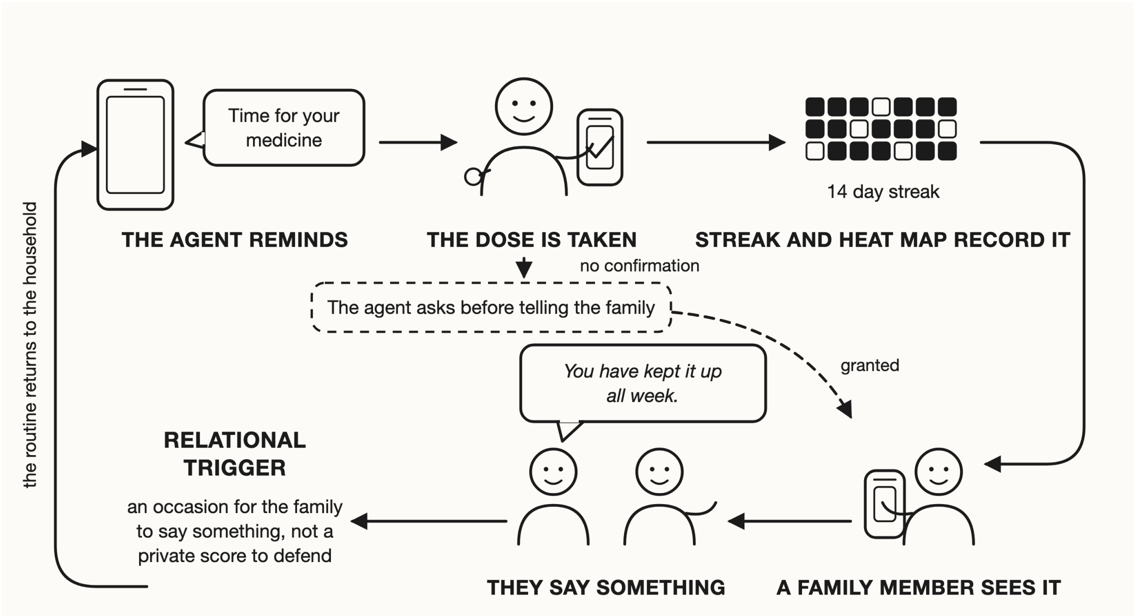


Fig. 2. Loop diagram: the agent reminds, the dose is taken, a streak and heat map record it, a family member sees the record and says something, and the routine returns to the household. A dashed branch shows an unconfirmed dose reaching the family only after the agent asks and the older adult grants.

Participants or family members therefore entered medication schedules manually. At scheduled times, the deployed prototype delivered reminders containing the medicine name and timing. Participants could confirm that a dose had been taken, which updated the medication logbook, streak, and daily heat map.

When a scheduled dose remained unconfirmed, the system could ask whether a family member should be notified. Participants could decline the request, and the family-notification path proceeded only after permission was granted. Medication schedules and changes in whom the system served remained under human control.

The streak and heat map were designed to make medication records visible within the household and create possible occasions for family response. This was a design rationale rather than an assumed effect of the system.

Figure 2 presents the interaction loop implemented during deployment. After a scheduled reminder, an older adult could confirm taking the dose, which updated the medication record, streak, and heat map. When a dose remained unconfirmed, the family-notification path could proceed only after the participant granted permission.

Four mechanisms in the broader design specification were not implemented in the deployed version: a risk-graded weakening veto, patterned interpretation of silence, a probationary onboarding mode, and shared family scores. These mechanisms were discussed during Phase 2 as proposed designs rather than as features participants had used.

### 3.5 Phase 2: Household Deployment and Post-Deployment Study

*3.5.1 Prototype Deployment* Researchers installed the prototype during a home visit and configured the initial medication schedule with the older adult or, where relevant, the family member operating the household device. Onboarding covered scheduled reminders, the logbook, and the spoken announcement identifying whom the system was serving.

Nine older adults used personal smartphones, while five used household smartphones shared with or operated by relatives. Device information was unavailable for three participants. We treated shared and intermediated operation as part of the deployment context rather than as equivalent to independent smartphone use, consistent with prior accounts from the region [4, 33, 98].

Participants used the prototype for two weeks before the Phase 2 interviews.

*3.5.2 Post-Deployment Interviews* We returned to participating households after deployment and conducted post-deployment interviews in Bangla. We used separate semi-structured protocols for older adults and family caregivers; the complete protocols are provided in **Appendix B1** and **Appendix B2**, respectively. When an older adult and an enrolled caregiver both participated, each discussed the deployment period from their own perspective.

Questions about deployed features covered reminders, announcements of whom the system was serving, medication records, requests for family involvement, occasions when participants accepted or declined those requests, family notifications, and changes participants noticed in their medication routines or family involvement. Interviewers first invited participants to describe experiences in their own terms and then used follow-up probes to reconstruct specific episodes.

The interviews also presented the four mechanisms that were not implemented. These were introduced as hypothetical design possibilities rather than experiences participants were assumed to have had. Several probes described possible interpretations of these mechanisms, so we distinguished episodes or interpretations volunteered by participants from agreement expressed after a probe introduced a possible interpretation. Prompted agreement alone was treated as weaker evidence during analysis.

Accounts of deployed features support claims about participants' reported experiences during the two-week period. Responses to unimplemented mechanisms support only claims about how participants reasoned about the proposed designs.

### 3.6 Data Analysis

*3.6.1 Transcription and Translation* The verified Bangla transcripts served as the primary analytic record. Automatic speech recognition produced preliminary Bangla transcripts, and machine translation produced initial English renderings, but neither automated output was treated as final. Two Bangla-fluent researchers verified each transcript against the corresponding audio and checked the English rendering against the verified Bangla source. Coding and interpretation were conducted using the Bangla records, and researchers returned to the original Bangla whenever kinship terms, honorifics, relational expressions, or other meanings were not adequately represented in English. Automated systems were used only to support preliminary transcription and translation, not coding, theme development, or interpretation.

*3.6.2 Reflexive Thematic Analysis* We analysed material from both phases using reflexive thematic analysis [15, 16]. Coding began inductively and attended to participants' accounts of medication routines, family involvement, responsibility, technology use, and experiences with the prototype. We worked across semantic and latent levels where relevant and treated themes as researcher-constructed interpretations rather than entities that independently emerged from the data [15, 16].

Analysis began with repeated reading and memoing. The two researchers who conducted the interviews developed preliminary codes and discussed interpretations iteratively with the supervising researcher. We did not calculate inter-rater reliability because coder agreement was not used as a criterion for theme validity within our reflexive approach [16, 93]. Researchers then grouped related codes into candidate themes and repeatedly examined them against the underlying transcripts. Themes were revised, combined, separated, or renamed as the analysis developed.

We retained older adults' and caregivers' accounts separately during analysis. When participants described the same household event differently, we treated those differences as part of the data rather than resolving them into a single household account. We maintained analytic memos, coding records, theme maps, and records of cases that complicated developing interpretations. When an interpretation depended on language not adequately represented in English, we returned to the verified Bangla transcript. We make no claim of thematic saturation.

### 3.7 Ethical Considerations

The study received approval from the Brac University Ethics Committee before data collection began. All study procedures followed the approved protocol.

Researchers explained the study purpose, procedures, voluntary nature of participation, audio recording, data use, and automated transcription and translation before obtaining informed consent. Study information was read aloud so participants who could not read received the same information. Consent was obtained individually, and separate permission was requested for audio recording. Caregivers consented independently from the older adults whose medication routines they might discuss.

Personal names were replaced with participant identifiers, and the identifier key was stored separately from the analytic corpus. Participants could discontinue participation at any point.

Three researchers conducted the study. Two conducted the interviews, transcript verification, and coding, while a third supervised the analysis. All three are fluent in Bangla and English. Recruitment began through the research team's social networks, which made some households more reachable than others [75, 103]. We therefore do not make population-level claims from the sample.

The two-week deployment supports accounts of participants' experiences during that period rather than longer-term use. We did not measure medication adherence, medication errors, or health outcomes and make no claims that the prototype changed them. Responses to unimplemented mechanisms are similarly limited to participants' judgments about described designs rather than observed use.

## References


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# A Phase 1 Interview Protocols

Phase 1 used separate semi-structured interview guides for older adults and family caregivers. Interviews were combined with situated observation and medication-routine walkthroughs in participants' homes. Interviewers used the questions below flexibly and followed relevant responses with probes about recent or specific episodes. Where appropriate, participants were asked to demonstrate medication practices, artifacts, and technology use rather than relying only on general descriptions.

## A.1 Appendix A1: Older-Adult Interview Protocol

*A.1.1 Phase 1: Environment and Artifact Observation* **Observational focus:** Before beginning the interview, document the immediate medication environment, including medicine-storage locations, lighting and noise conditions, physical accessibility, and visible aids such as prescriptions, written notes, calendars, pillboxes, or other organizational strategies.

*A.1.2 Phase 2: Background and Medication Routine*

(1) **Can you tell me a little about yourself and how medicines are part of your everyday routine?**
*Probe: How many medicines do you usually take, and at what times of day?*

(2) **Can you walk me through how you normally take your medicines during a typical day?**
*Probe: What happens from the time you realize a medicine is due until you take it?*

(3) **How do you know which medicine to take and when?**
*Probe: Do you remember it yourself, check a prescription or written note, look at the medicine strip, or ask someone?*

(4) **Who usually takes responsibility for different parts of your medication routine?**
*Probe: Which parts do you normally manage yourself, and which parts does someone else help with?*

(5) **Has this arrangement always been the same?**
*Probe: How did family members become involved in the parts they now handle?*

*A.1.3 Phase 3: Situated Medication Walkthrough* **Observational focus:** Ask the participant to demonstrate a usual medication routine. Observe how medicines are located, identified, handled, and organized and what artifacts or workarounds support the process.

(6) **Could you show me how you would prepare and take one of your usual doses?**
*Probe: How do you identify the correct medicine and dose?*

(7) **Where do you usually keep your medicines? Why do you keep them there?**
*Probe: Does anyone else organize, move, or prepare them for you?*

(8) **Do you use anything to help you remember or organize your medicines?**
*Probe: Written notes, phone alarms, pillboxes, marked medicine strips, particular storage locations, or something else?*

(9) **Have you tried any method that you later stopped using?**
*Probe: What did you try, and why did you stop?*

(10) **What parts of your current routine work well, and what parts are difficult?**
*Probe: If you could change one part of the routine, what would you change?*

*A.1.4 Phase 4: Disruptions, Missed Doses, and Handover* **Observational focus:** Ask about specific recent episodes rather than only general difficulties. Attend to changes in routine, location, family availability, and other circumstances surrounding the episode.

(11) **Have you ever missed or delayed a dose?**
*Probe: Tell me about the most recent time. What was happening that day?*

(12) **How did you realize that the dose had been missed or delayed?**
*Probe: Did you notice yourself, or did someone else notice or remind you? What happened afterward?*
(13) **Are there situations when taking your medicines becomes more difficult than usual?**
*Probe: What happens when you are away from home, have visitors, attend a social occasion, sleep at a different time, or your usual schedule changes?*
(14) **What happens when the person who usually helps with your medicines is unavailable?**
*Probe: Tell me about the last time this happened. Did you manage differently yourself, or did someone else become involved?*
(15) **Have you ever taken the wrong medicine or taken a medicine twice by mistake?**
*Probe: How was it noticed, and what happened afterward?*

*A.1.5 Phase 5: Prescription, Medicine Information, and Accessibility* **Observational focus:** Where appropriate, invite participants to handle their own prescription, medicine strip, bottle, or other medication information while discussing how they use it.

(16) **How do you usually understand what is written on your prescription or medicine packaging?**
*Probe: Which parts can you read or recognize yourself?*
(17) **What do you do when something about a medicine or prescription is unclear?**
*Probe: Whom do you usually ask: a family member, doctor, pharmacist, or someone else?*
(18) **When a new medicine is prescribed, how do you learn when and how to take it?**
*Probe: Who usually explains it to you?*
(19) **Is anything about reading medicine labels, prescriptions, or instructions difficult for you?**
*Probe: Small text, medicine names, dosage instructions, language, or identifying similar-looking medicines?*

*A.1.6 Phase 6: Technology and Device Use* **Observational focus:** Where participants use a digital device, invite them to demonstrate relevant interactions. Observe shared or intermediated use rather than assuming that device ownership means independent use.

(20) **What kind of phone do you usually use?**
*Probe: Is it your own phone or a phone shared with other household members?*
(21) **What do you normally use your phone for?**
*Probe: Who helped you learn to use it or set it up?*
(22) **Do you currently use a phone, alarm, application, or other digital tool to help with medicines?**
*Probe: Could you show me how you use it?*
(23) **What feels easy or difficult when you use a phone or other technology?**
*Probe: Reading the screen, hearing alerts, typing, navigating menus, remembering steps, or something else?*
(24) **Does anyone in your family sometimes operate or configure a phone or application for you?**
*Probe: What do they usually help with?*
(25) **What would make a medication reminder or tracking tool easier for you to use?**
*Probe: Larger text, spoken information, simpler controls, or another form of support?*

*A.1.7 Phase 7: Family Involvement and Medication Responsibility*

(26) **Who in your family is involved in your medicines?**
*Probe: What does each person usually do?*
(27) **How do family members know whether you have taken your medicine?**

*Probe: Do they ask you, watch the routine, look at the medicines, call you, or use some other way?*

(28) **What usually happens when someone reminds you about a medicine?**
*Probe: Can you tell me about a recent example?*

(29) **Are there parts of your medication routine that you prefer to manage yourself?**
*Probe: Which parts, and what makes those important for you to manage?*

(30) **When something about your medicines needs to change, how is that usually decided?**
*Probe: Who is involved in the discussion, and who normally acts on the change?*

(31) **Have you and a family member ever had different views about something related to your medicines?**
*Probe: Can you tell me about a specific occasion and what happened?*

(32) **How do you communicate with doctors or pharmacies about your medicines?**
*Probe: Do you usually communicate with them yourself, together with someone, or through another family member?*

#### *A.1.8 Phase 8: Closing and Design Reflection*

(33) **If something could help with one part of your medication routine, which part would you most want help with?**
*Probe: Which parts would you prefer to continue managing yourself?*

(34) **If a reminder system could not tell whether you had taken a medicine, what do you think it should do next?**
*Probe: Should it remind you again, wait, or involve someone else?*

(35) **If someone else needed to be informed about a missed medicine, who should that person be?**
*Probe: How should that decision be made?*

(36) **Who should be able to change the settings of a medication-support system?**
*Probe: Should you know when someone else makes a change?*

(37) **Is there anything a medication-support system should never do?**

(38) **Is there anything about managing your medicines that we have not talked about but that you think is important for us to understand?**

### A.2 Appendix A2: Family-Caregiver Interview Protocol

The caregiver was interviewed as a participant in their own right. Questions focused on the caregiver's own work, judgments, and position within household medication routines. When relevant, interviewers distinguished between the caregiver's own experience, their account of the older adult's experience, and events they described as occurring jointly.

*A.2.1 Phase 1: Environment and Artifact Observation* **Observational focus:** Document where medicines are stored, whether these spaces are managed primarily by the caregiver, the older adult, or jointly, and what medication aids are currently used or visibly abandoned. Note relevant prescriptions, written labels, trays, pillboxes, devices, and shared-device arrangements.

#### *A.2.2 Phase 2: Background and Position in the Care Network*

(1) **Can you tell me a little about yourself and who lives in this household?**
*Probe: Who in the household takes medicines every day, how many medicines do they take, and at what times?*

(2) **Which parts of medication management do you usually handle?**
*Probe: Walk me through the process from obtaining the medicine to the point when it is taken. Which parts are usually yours?*

(3) **How did these responsibilities come to be yours?**

*Probe: Was this discussed explicitly, or did the arrangement develop over time?*

(4) **Who else in the family takes part in this work?**
*Probe: What does each person do? Does that division remain the same or change across situations?*

*A.2.3 Phase 3: Situated Walkthrough and Handover* **Observational focus:** Request a physical walkthrough of a usual dose. Observe preparation, medicine identification, label reading, organization, and any improvised practices.

(5) **Could you show me how a typical dose is prepared and taken in this household?**
*Probe: How do you know which medicine is due and when? Show me where that information comes from.*

(6) **Can the older adult find and identify their medicines without your help?**
*Probe: Which medicines, and how do they recognize them?*

(7) **Have you used a pillbox, chart, written list, phone alarm, or another system to organize the medicines?**
*Probe: What happened when you used it? If you stopped, why?*

(8) **If you had to be away, how would this medication work be handed over to someone else?**
*Probe: Tell me about the last time this happened. What was difficult to communicate or coordinate?*

*A.2.4 Phase 4: Disruptions, Absence, and Missed Doses* **Observational focus:** Ground questions in recent episodes and identify where the caregiver was and what other household circumstances were present.

(9) **Has a dose ever been missed or substantially delayed?**
*Probe: Tell me about the most recent time. What was happening that day, and where were you?*

(10) **How did you find out that the dose had been missed, and what did you do?**
*Probe: Did you contact the older adult or someone else? What happened afterward?*

(11) **Has a missed dose ever led to a situation that concerned you?**
*Probe: Could you walk me through what happened?*

(12) **When is medication management most difficult for you?**
*Probe: What happens during work hours, travel, disrupted sleep, guests, social occasions, or religious observance?*

(13) **Has the older adult ever taken the wrong medicine or taken one twice?**
*Probe: How was this discovered, and what happened next?*

*A.2.5 Phase 5: Prescription Interpretation and Medicine Knowledge*

(14) **Who usually reads and interprets prescriptions in this household?**
*Probe: Which parts can the older adult manage independently, and which parts require help?*

(15) **What do you do when something on a prescription is unclear?**
*Probe: Tell me about the last time. Did you contact a doctor, pharmacist, or another family member?*

(16) **When a new medicine is added, how do other people in the household learn about it?**
*Probe: What information do you usually explain: the medicine name, timing, purpose, or something else?*

(17) **Has a medicine ever caused a reaction or side effect that concerned you?**
*Probe: How was it identified, and whom did you consult?*

*A.2.6 Phase 6: Technology, Proxy Use, and Device Sharing* **Observational focus:** Where appropriate, ask the caregiver to demonstrate relevant devices or tools. Attend to who owns, holds, configures, and operates each device.

(18) **What phones or other relevant devices are used in this household, and whose are they?**
*Probe: Are any of the devices shared?*

(19) **What does the older adult usually use their phone for, and how did they learn to use it?**

(20) **Who usually sets up applications, alarms, or settings on household members' phones?**

*Probe: Tell me about the last thing you configured for someone else.*

(21) **Have you ever set something up for the older adult that they later stopped using, turned off, or ignored?**
*Probe: What happened?*

(22) **Are there difficulties with reading medicine information, small text, or operating the device?**
*Probe: Could you show me an example using the medicine or device you normally use?*

#### A.2.7 Phase 7: Checking, Visibility, and Decision-Making

(23) **On an ordinary day, how do you know whether the older adult has taken their medicines?**
*Probe: Do you ask, observe, check the medicine, count remaining doses, or rely on another method? Tell me about yesterday.*

(24) **Does the older adult usually know when you are checking?**
*Probe: How does that checking normally happen?*

(25) **How does the older adult respond when you ask whether a medicine has been taken?**
*Probe: Can you tell me about a recent occasion?*

(26) **What is it like for you when you do not know whether a dose has been taken?**
*Probe: What do you usually do in that situation?*

(27) **When something about the medicines needs to change, how is the decision usually made?**
*Probe: Tell me about the most recent change. Who raised it, who discussed it, and who acted on it?*

(28) **Has the older adult ever changed something about the medication routine without telling you, or have you ever made a decision they did not initially agree with?**
*Probe: What happened afterward?*

(29) **Do relatives, friends, neighbours, or others outside the immediate care arrangement give medication advice?**
*Probe: Tell me about a specific occasion and what happened with that advice.*

#### A.2.8 Phase 8: Closing and Design Reflection

(30) **If something could take one part of this medication work off you, which part would you most want to give up?**
*Probe: Which part would you prefer to continue handling yourself?*

(31) **If a medication reminder system existed, who should it communicate with, and in what order?**
*Probe: Should it communicate differently with the older adult and with you?*

(32) **If a scheduled dose was not confirmed, what should happen next?**
*Probe: How long should the system wait? Should another person be informed?*

(33) **Who should be able to change the system's settings?**
*Probe: If the older adult changed something, would you want to know? If you changed something, should the older adult know?*

(34) **Is there anything such a system should never do in this household?**

(35) **Is there anything about looking after the older adult's medicines that we have not discussed but that you think is important?**
*Probe: What would you want people designing medication-support technology to understand about this work?*

## B Phase 2 Post-Deployment Interview Protocols

Phase 2 used separate semi-structured interview guides for older adults and family caregivers after the two-week prototype deployment. Interviewers first asked participants to describe their experiences in their own terms and then used probes to reconstruct specific episodes. Questions about features participants had used were distinguished from questions about design mechanisms that had not been implemented. Unimplemented mechanisms were introduced hypothetically and were treated as design judgments rather than accounts of use.

### B.1 Appendix B1: Older-Adult Post-Deployment Interview Protocol

#### *B.1.1 Section 1: Overall Experience and Everyday Use*

(1) **Can you tell me how using the system went for you over the past two weeks?**
*Probe: What became part of your usual routine? Probe: Was there anything you particularly liked, disliked, or stopped paying attention to?*

(2) **Can you walk me through a recent day when you used the system, from the first reminder onward?**
*Probe: Where were you? What did the system do? What did you do next?*

(3) **Did using the system change anything about how you managed your medicines?**
*Probe: What, if anything, did you do differently from before?*

(4) **Were there times when the system did not fit what was happening that day?**
*Probe: Tell me about the most recent time.*

#### *B.1.2 Section 2: Reminders, Confirmation, and Family Involvement*

(5) **What usually happened when a medicine reminder appeared?**
*Probe: What did you normally do after hearing or seeing it?*

(6) **Were there occasions when you did not confirm a dose?**
*Probe: Tell me about a specific occasion. Why was it left unconfirmed?*

(7) **What happened when a dose remained unconfirmed?**
*Probe: What did the system do next?*

(8) **Did the system ever ask whether a family member should be involved?**
*Probe: Tell me about the most recent time you remember.*

(9) **How did you decide whether to agree or decline when it asked?**
*Probe: What was happening in that situation?*

(10) **Were there situations when you preferred to handle the matter yourself?**
*Probe: What made those situations different?*

(11) **Were there situations when involving a family member was useful to you?**
*Probe: What made their involvement useful in that case?*

(12) **After a family member became involved, what happened next?**
*Probe: Did they call, ask you something, come to you, or respond in another way?*

#### *B.1.3 Section 3: Who the System Was Serving*

(13) **Did the system ever say who it was serving or working with?**
*Probe: What do you remember it saying?*

(14) **What did that announcement mean to you, if anything?**
*Probe: Did you usually notice it, or did it become part of the background?*

(15) **Did there seem to be times when the system was mainly helping you and other times when it was bringing your family into the situation?**
*Probe: Can you describe one of each?*
(16) **Was there ever a time when you wanted the system to involve someone differently from what it did?**
*Probe: What would you have preferred?*
(17) **Did you ever want to stop, delay, or change what the system was about to do?**
*Probe: What happened?*

#### B.1.4 Section 4: Family Checking and Medication Records

(18) **Before using the system, how did your family usually know whether you had taken your medicines?**
(19) **During the two weeks, did anything change about how family members checked or asked about your medicines?**
*Probe: More often, less often, or in a different way?*
(20) **How did you feel about the way family members used the information from the system?**
*Probe: Can you give me a specific example?*
(21) **Did the medication record, logbook, streak, or heat map ever come up in a conversation with someone in your family?**
*Probe: Who brought it up, and what was said?*
(22) **Did you pay attention to the streak or medication record yourself?**
*Probe: Did it affect anything you did?*
(23) **Who do you think should be able to see medication records like these?**
*Probe: Should that remain the same all the time, or should it be possible to change?*
(24) **Was there ever a difference between what the system recorded and what you or someone in your family believed had happened?**
*Probe: How was that handled?*

#### B.1.5 Section 5: Trust, Reliance, and Breakdown

(25) **Think back to the first few days of using the system. How did you decide whether you could rely on it?**
*Probe: Did you continue checking anything yourself?*
(26) **Did your way of using the system change as the two weeks went on?**
*Probe: What led to that change?*
(27) **Did the system ever get something wrong or behave differently from what you expected?**
*Probe: What did you do afterward?*
(28) **If the system stopped working tomorrow, what would you do about your medicines?**
*Probe: Would anything be difficult to return to?*
(29) **Do you think using the system changed how much you relied on your own memory, family members, or other reminders?**
*Probe: In what way?*

*B.1.6 Section 6: Proposed Design Mechanisms* *The mechanisms in this section were described as possibilities rather than features participants had used.*

(30) **Suppose you told the system not to involve a family member, but it believed the situation was becoming more concerning. What should it do?**
*Probe: Should your refusal always remain final, or are there circumstances when it should ask again?*

(31) **If it asked again, how should it explain why it was doing so?**

(32) **Suppose you did not respond to a reminder at all. What should the system assume?**
*Probe: Are there ordinary reasons you might not respond?*

(33) **How could the system distinguish an ordinary non-response from a situation that may need attention?**

(34) **Imagine that when you first started using the system, it stayed in a more limited mode until you became comfortable with it. Would that be useful?**
*Probe: What would you want it to do during that period? Probe: Who should decide when that period ends?*

(35) **Imagine that family members could also have shared scores or progress information connected to medication support. What would you think about that?**
*Probe: What, if anything, should be shared? Probe: Who should decide whether it is visible?*

#### B.1.7 Section 7: Closing Reflection

(36) **Taking everything together, who did the system seem to be working for during these two weeks?**
*Probe: Did that seem the same in every situation?*

(37) **Can you describe a situation in which you and a family member might want different things from the system?**
*Probe: What should the system do in that situation?*

(38) **When, if ever, should the system move from helping you privately to involving someone else?**
*Probe: Who should decide that?*

(39) **Can you take me through one recent interaction with the system from beginning to end?**
*Probe: What happened before it started, what did the system do first, what happened next, who became involved, and how did it end?*

(40) **Would you want to continue using a system like this?**
*Probe: Under what conditions?*

(41) **If you could change one thing about the system, what would you change?**

(42) **Is there anything about your experience that we have not discussed but that you think is important?**

### B.2 Appendix B2: Family-Caregiver Post-Deployment Interview Protocol

#### B.2.1 Section 1: Overall Experience and Household Use

(1) **Can you tell me how the past two weeks with the system went from your perspective?**
*Probe: What, if anything, changed in your usual medication-related work?*

(2) **Can you walk me through a recent occasion when the system became relevant to you?**
*Probe: What happened first, and how did you become involved?*

(3) **Was there anything about the system that was useful for you?**
*Probe: Can you describe a specific occasion?*

(4) **Was there anything that did not fit how medication care normally works in your household?**
*Probe: Tell me about the most recent example.*

#### B.2.2 Section 2: Reminders, Family Involvement, and Permission

(5) **When a dose was due, what did you normally know about it?**

*Probe: Did the system contact you directly, or did you usually learn about it another way?*

(6) **Did the system ever involve you after a dose remained unconfirmed?**
*Probe: Tell me about a specific occasion.*

(7) **What happened once you were notified?**
*Probe: What did you do, and how did the older adult respond?*

(8) **Were there occasions when you expected to be involved but were not?**
*Probe: What made you expect involvement in that situation?*

(9) **Did you know whether the older adult could decline involving you?**
*Probe: What did you think about that arrangement?*

(10) **Were there situations when you thought the older adult should handle the matter without you?**
*Probe: What made those situations different?*

(11) **Were there situations when you thought family involvement was particularly important?**
*Probe: Why?*

*B.2.3 Section 3: Who the System Was Serving*

(12) **Did the system ever say who it was serving or working with?**
*Probe: What do you remember it saying?*

(13) **What did you make of those announcements?**
*Probe: Did they affect how you understood your own role?*

(14) **Did the way the system involved you or the older adult seem to change across situations?**
*Probe: Tell me about a specific example.*

(15) **Was there a change you wanted in how the system involved family members that did not happen?**
*Probe: What would you have wanted instead?*

(16) **Was there anything the system told you that you were unsure the older adult knew had been shared?**
*Probe: What did you do in that situation?*

*B.2.4 Section 4: Checking, Communication, and Medication Records*

(17) **Before the deployment, how did you usually find out whether medicines had been taken?**

(18) **During the deployment, did anything change about how you checked?**
*Probe: Did you ask more, less, or differently?*

(19) **Did anything change about your conversations with the older adult concerning medicines?**
*Probe: Was there a kind of conversation you had before that happened less often during deployment?*

(20) **Did the older adult ever say anything about how you were checking or responding to information from the system?**
*Probe: What do you remember them saying?*

(21) **Did you use the logbook, streak, or heat map to understand what had happened?**
*Probe: How did you use it, if at all?*

(22) **Did anyone in the family talk about the streak or medication record?**
*Probe: Who raised it, and what happened afterward?*

(23) **Did the system's record ever differ from what you or another family member thought had happened?**
*Probe: How was the difference handled?*

(24) **Who do you think should be able to see medication records like these?**

*Probe: Who should decide, and should that decision be changeable?*

#### B.2.5 Section 5: Trust, Reliance, and Breakdown

(25) **Think back to the first few days. What did you do to decide whether you could rely on the system?**
*Probe: Did you continue checking anything independently?*

(26) **Did the way you relied on the system change during the two weeks?**
*Probe: What led to that change?*

(27) **Did the system ever get something wrong or behave differently from what you expected?**
*Probe: What did you do afterward?*

(28) **Do you still keep the older adult's medication schedule in your head in the same way as before?**
*Probe: Has anything changed?*

(29) **If the system stopped working tomorrow, what would happen in the household?**
*Probe: Walk me through how you would manage that day.*

#### B.2.6 Section 6: Proposed Design Mechanisms

*The mechanisms in this section were described as possibilities rather than features participants had used.*

(30) **Suppose the older adult declined family involvement, but the system believed the situation was becoming more concerning. What should happen next?**
*Probe: Should the system accept the decision, ask again, or do something else?*

(31) **Are there circumstances in which you think the system should continue asking even after someone has said no?**
*Probe: What limits should there be?*

(32) **Suppose nobody responded to a reminder. What should the system understand from that?**
*Probe: What ordinary situations could produce no response?*

(33) **How should it distinguish those situations from one that may need family attention?**

(34) **Imagine that during the first days of use, the system operated in a more limited mode while the household decided whether to trust it. Would that be useful?**
*Probe: What would you want to check during that period? Probe: Who should decide when the system can do more?*

(35) **Imagine that family members could share scores or progress information connected to medication support. What would you think about that?**
*Probe: What should be visible, to whom, and who should decide?*

#### B.2.7 Section 7: Closing Reflection

(36) **Taking everything together, who did the system seem to be working for during the deployment?**
*Probe: Did your answer change depending on the situation?*

(37) **Can you think of a situation in which what you wanted and what the older adult wanted were different?**
*Probe: What should the system do when that happens?*

(38) **When, if ever, should the system move from working directly with the older adult to involving the family?**
*Probe: Should it be able to make that change itself, or should someone authorize it?*

(39) **Can you take me through one recent interaction involving the system from beginning to end?**

*Probe: What happened beforehand, who was present, what did the system do, what did each person do, and how did it end?*

(40) **What do you think the older adult would say about that same interaction?**
*Record this separately from the caregiver's own account.*

(41) **Would you want the household to continue using a system like this?**
*Probe: Under what conditions?*

(42) **If you could change one thing about the system, what would you change?**

(43) **Is there anything about your experience or your medication-related work that we have not discussed but that you think is important?**